



GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

CURRICULUM DEVELOPMENT CELL

**PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING /
DIPLOMA IN INFORMATION TECHNOLOGY**

COURSE CODE: FC5462

COURSE TITLE: OPERATING SYSTEM

TEACHING SCHEME:

LEVEL OF COURSE	PRERE-QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V	-	03	-	02	05	16	48	-	32

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	25#	25^	150
	MIN.	28	---	40	10	10	

@: Internal Assessment #: External assessment-Practical based \$: online examination

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms. For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1. RATIONALE:

An Operating System is basically a system program that controls the execution of application programs and acts as an interface between applications and the computer hardware. It manages the computer system resources to be used in an efficient manner. This course enables to learn internal functioning of operating system and will help in identifying appropriate Operating System for given applications/task. This course is also a prerequisite for the group of courses included in 'Cloud Infrastructure Maintenance' Elective group.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

1. Install operating system and configure it.
2. Use operating system tools to perform various functions.
3. Execute process commands for performing process management operations.
4. Apply scheduling algorithms to calculate turnaround time and average waiting time.
5. Calculate efficiency of different memory management techniques.
6. Apply file management techniques.

3. DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topics and Subtopics	CO No.	Hrs	Marks
Unit 1. Overview of Operating System	1.a Explain the functioning of given component of OS. 1.b Explain characteristics of the given type of Operating system. 1.c Identify type of operating system suitable for the given type of application. 1.d Explain features of command line based OS and GUI based OS.	1.1 Operating system – Concept, Components of computer system , Operations of OS: Program management, Resource management , Security and protection; Views of OS: User view, System View 1.2 Different Types of Operating System- Batch operating system, multi programmed, Time Shared OS, Multiprocessor Systems, Distributed Systems, Real time systems. Mobile OS (Android, iOS). 1.3 Command line based OS-DOS, UNIX; GUI base OS-WINDOWS , LINUX	1	04	08
Unit 2. Services and components of the Operating System	2.a Start, stop, and restart the given Services in Linux. 2.b Explain use of the given System call of specified OS. 2.c Explain the process, OS follows in managing the given resources. 2.d Explain the uses of the given operating system .	2.1 Different Services of Operating System. 2.2 System Calls- Concept, types of system calls. 2.3 OS Components: - Process Management, Main Memory Management, File Management, I/O System management, Secondary Storage management. 2.4 Use of operating system to user management, security policy, device management, performance monitor, task scheduler	2	10	14

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topics and Subtopics	CO No.	Hrs	Marks
Unit 3. Process Management	3.a Explain various states of Process. 3.b Describe the components of PCB. 3.c Explain necessity of inter-process communication. 3.d Explain characteristics of executing the given multithreading model.	3.1 Process:- process states, Process Control Block (PCB). 3.2 Process Scheduling- Scheduling Queues, Schedulers, Context Switch. 3.3 Inter-process communication (IPC): Introduction, shared memory system and message passing system. 3.4 Threads - Benefits, users and kernel threads, Multithreading Models – Many to One, One to One, Many to Many.	3	10	12
Unit 4. CPU Scheduling and Algorithms	4.a Justify the need and objective of given job scheduling criteria with relevant example. 4.b Explain with example the procedure of allocating CPU to the given process using the specified OS. 4.c Calculate turnaround time and average waiting time of the given scheduling algorithm. 4.d Explain functioning of the given necessary condition leading to deadlock.	4.1 Scheduling types - scheduling Objectives, CPU and I/O burst cycles, Pre-emptive, Non- Pre-emptive Scheduling, Scheduling criteria. 4.2 Types of Scheduling algorithms – First come first served (FCFS), Shortest Job First (SJF), Shortest Remaining Time(SRTN), Round Robin (RR) Priority scheduling, multilevel queue scheduling. 4.3 Deadlock - System Models, Necessary Conditions leading to Deadlocks, Deadlock Handling - Preventions, avoidance.	4	08	12
Unit 5. Memory Management	5.a Describe the working of specified memory Management Techniques. 5.b Explain characteristic of the given memory management techniques. 5.c Write algorithm for the given page replacement technique. 5.d Calculate Page fault for the given page reference string.	5.1 Basic Memory Management - Partitioning, Fixed and Variable, Swapping, fragmentation. 5.2 Virtual Memory - Introduction to Paging, Segmentation and Page fault. 5.3 Page Replacement Algorithms: FIFO, LRU, Optimal.	5	08	12

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topics and Subtopics	CO No.	Hrs	Marks
Unit 6. File Management	6.a Explain structure of the given file system with example. 6.b Describe mechanism of the given file access method. 6.c Explain features of the given Raid level structure of hard disk.	6.1 File - Concepts, Attributes, Operations, types and File System Structure. 6.2 Access Methods - Sequential, Direct. File-Allocation Methods-Contiguous, Linked, Indexed. Free Space management Techniques-Bitmap and Linked List. 6.3 Directory structure-- Single level, two levels, tree structured directory, Disk Organization and disk Structure- Physical structure, Logical structure, Raid structure of disk, raid level 0 to 6.	6	08	12

4. LIST OF PRACTICALS:

Sr No.	PRACTICAL OUTCOMES (PrOs)	CO NO.
1	Install and configure Linux (or alike) operating system.	1
2	Execute general purpose commands date, time, cal, clear, banner, tty, script, man.	1
3	Work with multiple linux terminals and basic commands: who, who am I, login, passwd, su, pwd.	2
4	Use Operating services (Editor, GUI, File handling.) Run commands to start, stop, and restart the specified service in Linux.	2
5	Execute process commands- ps, wait, sleep, exit, kill.	3
6	Write a program to calculate total waiting and turnaround time of n processes with First Come First Serve CPU scheduling algorithm.	4
7	Write a 'C' program to implement FIFO page replacement algorithm.	5
8	Execute file and directory manipulation commands — ls, rm, mv, cp, join, split, cat (file saving and redirection operator), head, tail, touch,	6
9	Execute file and directory manipulation commands — diff, comm., pr, chmod, mkdir, rmdir, cd, pwd, dir, cmp. (Use wild card character).	6
10	Write and execute Shell Script by using following Control statements features- "if" statement	6
11	Write and execute Shell Script by using following Control statements features- "for" statement, exit, break, continue.	6
12	Write Shell script to find out whether -File has read, write, and execute permissions?	6

Note:

- The entire above listed practical's need to be performed compulsorily, so that the students reach the 'Precision level of Dave's Psychomotor Domain Taxonomy.
- The Process and Product related skills associated with each practical outcome shall be assessed on basis of following performance indicators.

Sr. No.	Performance Indicators	Weightage in %
a	Installation/configuration of operating system	25
b	Correctness of Executing various commands	25
c	Writing and executing programs to get desired output	20
d	Debugging the program	15
e	Submit journal report in time	15
	Total	100

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Follow safety practices.
- Practice good housekeeping.
- Demonstrate working as a leader/a team member.
- Maintain tools and equipment.
- Follow ethical Practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organizing Level' in 2nd year
- 'Characterizing Level' in 3rd year

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student- related co-curricular activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Prepare journal of practical's.
- Undertake micro-projects.

6. SUGGESTED INSTRUCTIONAL STRATEGIES

Following are suggested instructional strategies, which the teacher can adopt for the attainment of the various outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- The teacher needs to ensure to create opportunities and provisions for **co-curricular activities**.
- Guide student(s) in undertaking micro-projects.
- Demonstrate students thoroughly before they start doing the practice
- Use different instructional strategies in classroom teaching.
- Observe continuously and monitor the performance of student in lab.
- About 10-15% of the topics/sub-topics/contents which is relatively simpler or descriptive in nature may be given to the students for self-directed learning and assess the development of the COs through classroom presentations. Keep the record of the topics/sub-topics/contents given to the students.

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is to be undertaken by a student in the beginning of the semester. S/he ought to submit it by the end of semester. Each micro-project should include two or more Cos. The micro-project could be industry application based, internet based, laboratory based, field based and survey

based. In all semesters, it could be group-based (group 5-6 student) to build up skill and confidence in every student. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The student ought to submit micro-project by the end of the semester to develop the industry oriented COs.

A suggestive list of micro-projects is given here. Similar micro-projects could be added by the concerned faculty:

- a) Create a report depicting features of different types of Operating systems- Batch operating system, Multi Programmed, Time Shared, Multiprocessor Systems, Real time systems. , Mobile OS with example.
- b) Make a comparative statement to calculate page fault for given page reference string by using different page replacement algorithms.
- c) Prepare help guide using shell script for all the major Linux commands.
- d) Make a comparative chart to calculate total waiting and turnaround time of n processes
- e) with different CPU scheduling algorithm.
- f) Given with the P number of processes in the memory and N number of needed resources by them to complete their execution and the task is to find the minimum number of resources R which should be allotted to the processes such that deadlock will never occur.
- g) Implement first fit, best fit algorithm for memory management and write comparative statements.
- h) Make a comparative chart of various mechanisms of the given file access method.
- i) Make a comparative chart of directory structure and Raid level structure of hard disk.
- j) Make a comparative chart of threads and various multithreading models.
- k) Make a comparative chart of various deadlock avoidance and prevention algorithms
- l) Execute text processing tr, wc, cut, paste, spell, sort, grep, more.
- m) Use vi editor and perform all editor commands.
- n) Write Shell script to find out whether - Given file exists

Any 'other micro-projects suggested by subject faculty on similar line.
(Use features of `C* or shell scripts to develop above listed applications)

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

Sr No.	Equipment Name with Broad Specification	Practical No.
1	Computer system (Any computer system with basic configuration)	All
2	Linux or alike operating system such as Ubuntu, CentOS or any other.	

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Overview of Operating System	08	02	02	04
2	Services and Components of Operating System	10	02	04	04
3	Process Management	14	02	04	08
4	CPU Scheduling and Algorithms	14	02	04	08
5	Memory Management	14	02	04	08
6	File Management	10	02	04	04
Total		70	12	22	36

R= Remember, U= Understanding, A=Application and above (Bloom's Revised taxonomy)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

10. SUGGESTED LEARNING RESOURCES:

Sr. No.	Title Of Book	Author	Publication
1.	Operating System Concepts	Silberschatz, Galvin	John Wiley and Sons, Ninth Edition, 2015, ISBN: 978-51-265-5427-0
2.	Operating System	Godbole, Achyut S.	Tata McGraw Hill Education, 2015, ISBN: 9780070591134
3.	Operating Systems: Internals and Design Principles	Stallings, William	Pearsons, 8 edition 2015 ISBN: 978-0133805918
4.	Unix Concept and Programming	Das, Sumitabha	McGraw Hill education, 2015, ISBN: 978-0070635463
5	Operating System	Dhamdhare, Dhanjay M.	McGraw Hill, 2015 ISBN MO 978-1-25-900558-9
6	Operating System	Dr. Rajendra Kawalw	Devraj Publications, Mumbai ISBN 978-81-933551-1-4

11. SUGGESTED SOFTWARE/LEARNING WEBSITES

- www.cs.wisc.edu/~bart/537 lecture notes-University of Wisconsin Madison.
- www.cs.kent.edu/osf03/notes/index.html- Vilinius Gediminas Technical University
- <http://www.howstuffworks.com/operating-system1.htm>
- www.computerhope.com/jargon/o/os.htm
- www.en.wikipedia.org/wiki/Operating_system
- <https://www.cs.uic.edu/Hbell/CourseNotes/OperatingSystems/125/masstorage.html>

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1	Dr.P.R.Satav	Lecturer in Computer Engineering.	Government Polytechnic Amravati
2	Smt. A.P.Jane	Lecturer in Information Technology	Government Polytechnic Amravati

Government Polytechnic, Amravati Programme Board of Studies Computer Engineering has approved the above course curriculum on 14/05/2020 and is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 14/07/2020
The Governing Body has approved the above course curriculum on 13/08/2021